Chapter 1 Topological Spaces

1.1 Definitions

1.1.1 Topological Space

Definition 1.1.1.1 ...

1.1.2 Generating of Topological Spaces

1.1.3 Bases, Subbases

1.1.4 Continuous Mapping

1.2 Constructions

1.2.1 Quotient Space

Quotient Topology

Definition 1.2.1.1 Suppose X be a set with equivalence relation $(X, \sim)$:

- Given $x \in X$, we denote $[x]$ the unique equivalence class that contains x;
- We refer to the map

  $$p : X \to X/\sim \qquad x \mapsto [x]$$

  Natural projection;
- Let $f : X \to Y$ be a map between sets such that $f(x) = f(y)$ whenever $x \sim y$. By the quotient factorization the map

  $$g : X/\sim \to Y \qquad [x] \mapsto f(x)$$

  is well-defined and it is the unique map $g : X/\sim \to Y$ such that $f = g \circ p$.

♠ **Proposition 1.2.1.2** (Topological-Quotient Proposition)

◦ **Proof** ... □

Definition 1.2.1.3 ...

Open Map, Closed Map

1.2.2 Product, Coproduct

1.2.3 Limits and Colimits

1.2.4 Pullback,Pushout

Pullback

♠ **Proposition 1.2.4.1** Let $p : X \to B$ be a continuous map of topological spaces. Let C be a topological space and $f : C \to B$ be a continuous map. We consider the topological space:

$$f^*X := \{(c, x) \in C \times X \mid f(c) = p(x)\}$$

and the maps

$$\begin{aligned} \varphi : f^*X \to C \qquad (c, x) \mapsto c \\ \psi : f^*X \to X \qquad (c, x) \mapsto x \end{aligned}$$

The following statements hold:

- 1. The maps $\varphi : f^*X \to C$ and $\psi : f^*X \to X$ are continuous;
  2. Suppose we are given a topological space W and two continuous maps $\alpha : W \to C$ and $\beta : W \to X$ such that

     $$f \circ \alpha = p \circ \beta$$

     The map

     $$\begin{aligned} \Phi : W \to f^*X \\ w \mapsto (\alpha(w), \beta(w)) \end{aligned}$$

     is continuous. In particular we obtain the following commutative diagram

     $$\begin{tikzcd}
W \arrow[drr, bend left, "\beta"] \arrow[dr, dashed, "\exists!\Phi"] \arrow[ddr, bend right, "\alpha"'] & & \\
& f^*X \arrow[r, "\psi"] \arrow[d, "\varphi"] & X \arrow[d, "p"] \\
& C \arrow[r, "f"] & B
\end{tikzcd}$$

  A map $\Phi : W \to f^*X$ is continuous iff the maps $\varphi \circ \Phi : W \to C$ and $\psi \circ \Phi : W \to X$ are continuous.
- The topological space f^*X with natural maps $\varphi : f^*X \to C$ and $\psi : f^*X \to X$ is the pullback, in the category $\mathcal{T}op$.

∘ **Proof**

- – Consider f^*X as a subspace of $W = X \times C$. Then, φ and ψ can be seen as restriction of π_C, π_X on f^*X.
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□

♠ **Proposition 1.2.4.2** (Pullback-Properties Proposition) Let $p : X \to B$ be a continuous map of topological spaces, let C be a topological space and let $f : C \to B$ be a continuous map:

- If X and C are compact and if X, B, C are Hausdorff, then f^*X is compact;
- Connectedness:
  - Even if X, B, C are connected, it does not necessarily follow that the pullback f^*X is connected;
  - Suppose that X, B, C are connected. We also assume that for each $b \in B$ the preimage $p^{-1}(b) \subseteq X$ is connected. It does not follow f^*X is connected;
- If X, C are Hausdorff, the pullback f^*X is also Hausdorff.

∘ **Proof** ... □

Pushout

♠ **Proposition 1.2.4.3** (Topological-Pushout Proposition) Let $f : A \to Y$ and $g : A \to Z$ be two continuous maps between topological spaces. We consider the topological space

$$Y \cup_A Z := (Y \sqcup Z)/\sim \qquad f(a) \simeq g(a), a \in A$$

and the map

$$i : Y \to Y \sqcup_A Z \qquad j : Z \to Y \sqcup_A Z$$

The following statements hold:

- $U \subseteq Y \cup_A Z$ is open$\Leftrightarrow$ $i^{-1}(U) \subseteq Y, j^{-1}(U) \subseteq Z$ are open.
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∘ **Proof** ... □

Definition 1.2.4.4 ...

♠ **Proposition 1.2.4.5** (Pushout-Properties Proposition) Let $f : A \to Y, g : A \to Z$ be two continuous maps betwween topolgoical spaces. We consider the corresponding pushout

1.2.5 Fibre Product

1.2.6 Specialization

Definition 1.2.6.1 In topology, the **specialization preoreder** is a preorder on the point sets of topological space. For **T0 Space**, this is a partial order. called **specialization order**: For topological space (X, τ),

$$x \le y \Leftrightarrow x \in \overline{\{y\}}$$

♠ **Proposition 1.2.6.2** We have:

- If the space is T_0, the preorder is a partial order.
- The symmetry of specialization preorder: $x \leq y \land y \leq x \to x = y \Leftrightarrow R_0$ Axiom;
- For T_1 Space, we have

$$x \leq y \Leftrightarrow x = y$$

- For sober space with specialization order $(X, \leq)$:
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  -

∘ **Proof** ... □

Morphsims between Specialization Preorders

1.3 Locally Closed Space

♠ **Proposition 1.3.0.1** The following properties are equivalent:

- E is the intersection of an open set and a closed set in X;
- For each $x \in E$, there is a neighborhood U of x such that $E \cap U$ of x such that $E \cap U$ is closed in U;
- E is open in its closure $\bar{E}$;
- The set $\bar{E} - E$ is closed in X;
- E is the difference of two closed sets in X;
- E is difference of two open sets in X;

∘ **Proof**

- (1)$\Rightarrow$ (2)
- (2)$\Rightarrow$ (3)
- (3)$\Rightarrow$ (4)
- (4)$\Rightarrow$ (5)
- (5)$\Rightarrow$ (6)
- (6)$\Rightarrow$ (1)

□

1.4 Noetherian Space

Definition 1.4.0.1

- A topological space is called **Noetherian** if the descending chain condition holds for closed subsets of X.
- A topological space is called **locally Noetherian** if every point has a neighborhood which is Noetherian.

◇ **Lemma 1.4.0.2** Let X be a topological space:

- Any subset of X with the induced topology is Noetherian;
- The space X has finitely many irreducible components;
- Each irreducible component of X contains a nonempty open of X;

◦ **Proof**

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□

◇ **Lemma 1.4.0.3** (Noetherian Property via Continuous Mapping) Let $f : X \to Y$ be a continuous map of topological spaces:

- If X is Noetherian, then $f(X)$ is Noetherian;
- If X is locally Noetherian and f open, then $f(X)$ is locally Noetherian;

◦ **Proof** ... □

◇ **Lemma 1.4.0.4** Let X be a topological space and $X_i \subseteq X; i = 1, \dots, n$. If each X_i is Noetherian, then $\cup X_i$ is Noetherian(with induced topology).

◦ **Proof**

□

◇ **Lemma 1.4.0.5** Let X be a locally Noetherian topological space, then X is locally connected.

◦ **Proof** ... □

1.5 Elementary Knowledge in Dimension

1.5.1 Krull Dimension

Definition 1.5.1.1 Let X be a topological space:

- A chain of **irreducible closed subsets** of X is a sequence

$$Z_0 \subset \cdots \subset Z_n \subset X$$

with Z_i

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-
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1.5.2 Codimension and Catenary Maps

Definition 1.5.2.1 Let X be a topological space. Let $Y \subseteq X$ be an **irreducible** closed subset. The **codimension** of Y in X is the supremum of lenghts e of chains

$$Y = Y_0 \subset \cdots \subset Y_e \subset X$$

◇ **Lemma 1.5.2.2** Let X be a topological space, and $Y \subseteq X$ is an irreducible closed subset. And $U \subseteq X$ is an open subset such that $Y \cap U$ is nonempty, then

$$codim(Y, X) = codim(Y \cap U, U)$$

∘ **Proof** ... □

1.5.3 Catenary Space

Definition 1.5.3.1 ...

1.6 Jacobson Spaces

Definition 1.6.0.1 Let X be a topological space, and X_0 be the set of **closed points** in X(i.e. $\overline{\{x\}} = \{x\}$). We say that: X is **Jacobson** if every closed subset $Z \subseteq X$ is the closure of $Z \cap X_0$.

◇ **Lemma 1.6.0.2** A topological space X is Jacobson iff every locally closed subset of X has a point closed in X.

∘ **Proof** ... □

Remark

◇ **Lemma 1.6.0.3** Let X be a topological space and X_0 be a set of closed points in X. Suppose that for every $x \in U \cap Z$, we have that

$$...$$

∘ **Proof** ... □

1.7 Limit of Spces

Chapter 2 Filter,Nets

2.1 Convergence of Filters

2.1.1 Filter

Definition 2.1.1.1 Let E be a set: A family of subsets of E satisfies

- $\varnothing \notin \mathcal{F}$;(F1)
- $A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$;(F2)
- $B \in \mathcal{F}, B \subseteq A \Rightarrow A \in \mathcal{F}$;(F3)

♣ **Example 2.1.1.2** Consider a sequence of points $x_1, \dots, x_n \in E$:

Definition 2.1.1.3 The **associated filter** is the family of all subsets of E such that: The subset of E contains all $x_i, i \in \mathbb{N}$ except possibly a finite number of items.

Basis of Filter

Definition 2.1.1.4 A family $\mathfrak{B}$ is a **basis of filter** $\mathcal{F}$ if:

- $\mathfrak{B} \subseteq \mathcal{F}$;(BF1)
- Every subset of E belonging to $\mathcal{F}$ contains some subsets of E belongs to $\mathfrak{B}$:

$$S \in \mathcal{F} \Rightarrow \exists T \in \mathfrak{B}, T \subseteq S$$

  (BF2)

♣ **Example 2.1.1.5** Filter of neighborhoods of 0 on real line:

$$\mathcal{F} = \{(-a, a) | a > 0\}$$

♣ **Example 2.1.1.6** Consider a sequence with $S = \{x_i | i \in \mathbb{N}\}$, then the filter given by

$$S_n := \{x_n, x_{n+1}, \dots\}, S_0 \supseteq S_1 \supseteq \dots$$

is a basis of $\mathcal{F}$.

Question Given some family of subsets $\mathcal{A}$. Is there a filter $\mathcal{F}$ having basis $\mathcal{A}$ as a basis?

◇ **Lemma 2.1.1.7** Given two filter $\mathcal{F}_1, \mathcal{F}_2$ with basis of $\mathfrak{B}$. Then, $\mathcal{F}_1 = \mathcal{F}_2$. It is uniquely determined.

∘ **Proof** Suppose $S \in \mathcal{F}_1$, then there exists some $T \in \mathfrak{B}$ such that $T \subseteq S$. And then, $S \in \mathcal{F}_2$. □

So finally, this leads the question to: Is $\mathcal{F}$ a filter? (F2) equivalent to:

- The intersection of any two sets beloing to $\mathcal{A}$, contains a set which belongs to $\mathcal{A}$.(BF)

This is weaker than (F2). Thus we may state:

A basis of filter on E is a family of nonempty subsets of E satisfying condition (BF).

The filter is uniquely determined by (BF2)

Comparison of Filters

Consider the order relation induced by $(P(E), \subseteq)$:

- Finer;
- less fine;

Topology

The definition of topology can be rewritten as following:

- $S \in \mathcal{F}(x) \Rightarrow x \in S$;(N1)
- $U \in \mathcal{F}(x) \Rightarrow \exists V \in \mathcal{F}(x), \forall y \in V, S \in \mathcal{F}(y)$.(N2)

Such $\mathcal{F}(x)$ is called the **filter of neighborhoods** of the point x.

2.1.2 Convergence with Filter

Definition 2.1.2.1

- Define: Filter $\mathfrak{F}$ converges to the point $x \Leftrightarrow$ Every neighborhood of x belongs to $\mathfrak{F}$.
  In other word: $\mathcal{F}$ is finer filter than neighborhoods of x.
- Sequence (x_n) converges to x if its associated filter converges to x.

2.1.3 Continuity

Then: The continuity can be rewritten as the following form

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2.2 Moore-Smith Convergence

2.2.1 Directed Set, Nets

Definition 2.2.1.1 ...

Chapter 3 Connectedness and Irreducibility

3.1 Connectedness

3.2 Irreduciblity

Chapter 4 Compactness

4.1 Compactness

4.1.1 Compact Spaces

Definition 4.1.1.1

- We say: A space is **(quasi-)comapct** if every open covering $X = \cup U_i$ has a finite subcovering;
- A continuous map $f : X \to Y$ is **(quasi-)comapct** iff the inverse image $f^{-1}(V)$ is compact for any compact $V \subseteq Y$;
- A subset $Z \subseteq X$ is called **retrocompact** if for every compact open $U \subseteq X$, $Z \cap U$ is compact;

4.1.2 Compact Maps

4.2 Locally Compact Spaces

4.3 Constructible Sets

Definition 4.3.0.1 Let X be a topological space and $E \subseteq X$ is a subset:

- We say E is **constructible** in E iff E is a finite union of subsets of the form

$$U \cap V^C$$

  where U, V are both open and retrocompact;
- We say E is **locally constructible** if for there exists an open covering $X = \cup V_i$ such that each $E \cap V_i$ is constructible in V_i;

◇ **Lemma 4.3.0.2** The collection of constructible sets is closed under:

- Finite Union;
- Finite Intersection;
- Compelement;

◦ **Proof**

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□

4.4 Partition of Unity

Chapter 5　Separation Axioms

Chapter 6　Conuntable Axioms

Chapter 7 Function Spaces and Metrization Problem

Chapter 8 Baire Spaces

Chapter 9   Dimension